Grade 5 Mathematics

Մաթեմատիկա — 5-րդ դասարան

COURSE CODE	MTH-G5
PROGRAM	Elementary Mathematics · Տարրական մաթեմատիկա
LEVEL	Grade 5 · typical age 10–11
DURATION	72–84 instructional hours (approx. 66 lessons)
PREREQUISITE	Grade 4 Mathematics or equivalent: multi-digit multiplication and division, fraction equivalence, decimals to hundredths.
LEADS TO	Grade 6 Mathematics (MTH-G6)

COURSE DESCRIPTION

Դասընթացի նկարագրություն

Grade 5 completes elementary arithmetic. Students finish the operations with fractions — including addition and subtraction with unlike denominators, multiplication of fractions by fractions, and division involving unit fractions — and extend the four operations to decimals through thousandths. Volume is introduced as a three-dimensional measure, the coordinate plane appears for the first time, and expressions with grouping symbols prepare the ground for algebra. By June the student should be able to compute confidently with any rational number they will meet in middle school.

TEACHING APPROACH

Դասավանդման մոտեցումը Վարժարանում

Grade 5 is the last year in which arithmetic can be repaired cheaply. Every gap left open here becomes a ratio problem in Grade 6, a proportional-reasoning problem in Grade 7, and an algebra problem in Grade 8. Վարժարան therefore treats Grade 5 as a consolidation year with an unusually rigorous diagnostic posture: the teacher's default assumption is that a wrong answer signals a conceptual gap upstream, not carelessness, and lesson time is spent finding it. Fraction multiplication and division are taught with models and language ('two thirds *of* six'), never as inverted rules to memorize.

PLACEMENT AND READINESS INDICATORS

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The placement assessment checks each indicator below. A student missing two or more is enrolled with a remediation plan attached to the personalized curriculum.

READINESS INDICATOR	WHAT IT TELLS THE TEACHER
Multiplies two-digit by two-digit numbers accurately	If weak, revisit Grade 4 Unit 2 area model — decimal multiplication will otherwise be unreachable.

READINESS INDICATOR	WHAT IT TELLS THE TEACHER
Adds and subtracts fractions with like denominators and converts mixed numbers	Direct prerequisite for Unit 2; a common gap in transferring students.
Compares decimals to hundredths and explains why	Prerequisite for all decimal operations.
Divides four-digit by one-digit numbers with remainders	Prerequisite for two-digit divisors and for decimal division.

COURSE LEARNING OUTCOMES

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On completion of this course the student will be able to:

1. Understand the place-value system through thousandths, including powers of ten and their exponents.
2. Read, write, compare, and round decimals to thousandths.
3. Fluently multiply multi-digit whole numbers using the standard algorithm.
4. Divide up to four-digit dividends by two-digit divisors, explaining the reasoning.
5. Add, subtract, multiply, and divide decimals to hundredths, and explain the reasoning used.
6. Add and subtract fractions and mixed numbers with unlike denominators.
7. Multiply a fraction by a fraction and interpret the product as scaling (resizing).
8. Divide a unit fraction by a whole number and a whole number by a unit fraction.
9. Interpret a fraction as division of the numerator by the denominator.
10. Write and interpret numerical expressions using parentheses, brackets, and braces.
11. Generate two numerical patterns from rules, form ordered pairs, and graph them.
12. Understand volume as a three-dimensional measure and find volumes of right rectangular prisms.
13. Graph points in the first quadrant and classify two-dimensional figures in a hierarchy.

COURSE UNITS

Դասընթացի բաժիններ

Each unit below lists the lesson-level topics a teacher delivers, the objectives that define success, the observable mastery criteria used to update the student's progress profile, and the misconceptions this unit reliably produces together with the recommended teacher response.

UNIT 1 · 8–9 LESSONS

Place Value, Decimals, and Powers of Ten

Կարգային արժեք, տասնորդականներ և տասի աստիճաններ

TOPICS COVERED

- Decimals through thousandths in standard, word, and expanded form
- Each place as ten times the place to its right and one tenth of the place to its left
- Powers of ten and exponent notation
- Multiplying and dividing by powers of ten and the movement of the decimal point
- Comparing and ordering decimals to thousandths
- Rounding decimals to any place
- Fluent multi-digit whole-number multiplication
- Division with two-digit divisors using partial quotients
- Estimating quotients

LEARNING OBJECTIVES

1. Write a decimal to thousandths in expanded form using fractions and using powers of ten.
2. Explain why multiplying by 10 shifts digits one place left, in place-value terms.
3. Divide a four-digit number by a two-digit divisor and justify the quotient.
4. Round any decimal to a specified place.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Compares and orders decimals to thousandths at 90% accuracy.
- Divides by two-digit divisors at 85% accuracy with reasoning shown.
- Explains the effect of multiplying by a power of ten without saying 'move the decimal'.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
'Multiplying by ten adds a zero' — applied to decimals, giving 4.5 × 10 = 4.50.	This is why the rule must never be taught as 'add a zero'. The digits shift because each is now worth ten times as much. Use a place-value chart with sliding digits.
More digits after the decimal point is read as a larger number.	Compare 0.7 and 0.65 on a number line and with a thousandths grid. Annexing zeros to equalize digit counts also helps.

TEACHING NOTE

Powers of ten are introduced here with exponent notation. This is the student's first exponent — treat it as an event, and connect it explicitly to the ten-times rule from Grade 4.

UNIT 2 · 8-9 LESSONS

Adding and Subtracting Fractions with Unlike Denominators

Տարբեր հայտարարով կոտորակների գումարում և հանում

TOPICS COVERED

- Why unlike denominators cannot be added directly
- Finding common denominators with models
- The least common denominator and why any common denominator works
- Adding and subtracting fractions with unlike denominators
- Adding and subtracting mixed numbers with unlike denominators, including regrouping
- Estimating fraction sums with benchmarks 0, 1/2, and 1
- Multi-step fraction word problems
- Assessing reasonableness of a fraction answer

LEARNING OBJECTIVES

1. Explain with a model why 1/2 + 1/3 is not 2/5.
2. Add and subtract fractions and mixed numbers with unlike denominators accurately.
3. Estimate a fraction sum with benchmarks and use the estimate to catch errors.
4. Solve a two-step word problem involving fraction addition and subtraction.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Adds and subtracts unlike-denominator fractions at 90% accuracy.
- Handles mixed-number subtraction with regrouping in 4 of 5 trials.
- Produces a benchmark estimate before computing in 4 of 5 tasks.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Numerators and denominators are added separately.	The defining error of Grade 5 fractions. Return to the unit-fraction meaning and to fraction strips — fourths and thirds are different-sized pieces and cannot be counted together.
Only the least common denominator is believed to work.	Show that 1/2 + 1/3 gives the same result with denominator 6, 12, or 24. Understanding this frees the student from a lookup step and reduces anxiety.

TEACHING NOTE

Benchmark estimation is the single most effective self-checking habit in fraction work. Require it on every problem for the first four lessons until it becomes automatic.

UNIT 3 · 9-10 LESSONS

Multiplying and Dividing Fractions

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TOPICS COVERED

- A fraction as division: 3/4 as 3 ÷ 4
- Multiplying a fraction by a whole number, revisited
- Multiplying a fraction by a fraction with an area model
- The meaning of 'of' in fraction multiplication
- Multiplication as scaling: when does the product shrink?
- Multiplying mixed numbers
- Area of a rectangle with fractional side lengths
- Dividing a whole number by a unit fraction
- Dividing a unit fraction by a whole number
- Real-world division problems with fractions

LEARNING OBJECTIVES

1. Interpret 3/4 as 3 ÷ 4 and connect it to a sharing situation.
2. Multiply two fractions using an area model and explain the result.
3. Predict without computing whether a product will be larger or smaller than a given factor.
4. Model and solve 4 ÷ 1/2 and 1/2 ÷ 4, and explain why the answers differ.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Multiplies fractions and mixed numbers at 90% accuracy.
- Predicts the scaling effect correctly in 8 of 10 trials before computing.
- Models both unit-fraction division types correctly in 4 of 5 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
'Multiplication makes bigger; division makes smaller' persists from Grade 3.	Directly confront it. Compute 6 × 1/2 and 6 ÷ 1/2 side by side and ask the student to explain the surprise. This is a genuine conceptual shift and deserves a full lesson.
Common denominators are found before multiplying fractions.	Over-generalization from Unit 2. Use the area model to show why multiplication needs no common denominator — the two operations do genuinely different things.
4 ÷ 1/2 is answered as 2.	Ask 'how many halves fit in 4?' and model with fraction strips. The measurement interpretation of division makes this immediate.

TEACHING NOTE

Never teach 'keep-change-flip' this year. Grade 6 formalizes fraction division; Grade 5's job is to build the meaning that makes the Grade 6 rule sensible rather than magical.

UNIT 4 · 8-9 LESSONS

Decimal Operations

Գործողություններ տասնորդական կոտորակների հետ

TOPICS COVERED

- Adding and subtracting decimals with place-value alignment
- Estimating decimal sums and differences
- Multiplying a decimal by a whole number
- Multiplying two decimals and reasoning about the decimal point
- Why the number of decimal places in a product is what it is
- Dividing a decimal by a whole number
- Dividing by a decimal using equivalent expressions
- Relating decimal operations to fraction operations
- Money and measurement problems with decimals

LEARNING OBJECTIVES

1. Add and subtract decimals with correct alignment and explain why alignment is required.
2. Multiply two decimals and justify the placement of the decimal point using fractions or estimation.
3. Divide a decimal by a whole number and by a decimal.
4. Solve multi-step problems involving money and measurement.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Computes all four decimal operations at 90% accuracy.
- Justifies decimal-point placement by estimation or by fraction equivalence, not by a counting rule.
- Solves multi-step decimal word problems correctly in 8 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Decimals are added right-aligned like whole numbers: 3.5 + 0.42 = 3.92 becomes 3.47.	Align by place value, not by the right edge. Annex zeros so both numbers show the same places; require this in writing.
The decimal-point rule for products is memorized and then misapplied.	Estimate first: 4.2 × 3.1 is about 12, so the answer cannot be 1.302. Estimation makes the rule self-correcting.

TEACHING NOTE

Every decimal operation should be connected back to its fraction equivalent at least once. 0.3 × 0.4 is 3/10 × 4/10 = 12/100 — this is where the placement rule comes from, and students who see it once rarely misplace the point again.

UNIT 5 · 6-7 LESSONS

Volume and Measurement

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TOPICS COVERED

- Volume as a count of unit cubes
- Measuring volume by packing and by counting layers
- The formulas $V = l \times w \times h$ and $V = B \times h$, and why they agree
- Volume of solid figures composed of two right rectangular prisms
- Converting among units within a measurement system
- Multi-step measurement conversion problems
- Line plots with fractional data and problems drawn from them

LEARNING OBJECTIVES

1. Find the volume of a right rectangular prism by packing, by layering, and by formula, and show the three agree.
2. Decompose a composite solid into prisms and find the total volume.
3. Convert between units within a system in a multi-step problem.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Computes volume correctly with cubic units in 9 of 10 trials.
- Finds the volume of a composite figure in 4 of 5 trials.
- Performs multi-step conversions correctly in 8 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Volume is computed by adding the three dimensions.	Return to packing with unit cubes. The layer model — area of the base times the number of layers — makes multiplication necessary rather than arbitrary.
Cubic units are omitted or written as square units.	Require units on every answer; treat wrong units as a wrong answer.

TEACHING NOTE

Introduce $V = B \times h$ alongside $V = l \times w \times h$. The $B \times h$ form generalizes to cylinders and prisms in middle school; the $l \times w \times h$ form does not.

UNIT 6 · 7–8 LESSONS

Expressions, Patterns, and the Coordinate Plane

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TOPICS COVERED

- Writing and evaluating numerical expressions with parentheses, brackets, and braces
- Order of operations with grouping symbols
- Interpreting an expression without evaluating it
- Generating two numerical patterns from two rules
- Forming ordered pairs from corresponding terms
- The coordinate plane: axes, origin, and ordered pairs in the first quadrant
- Graphing points and interpreting coordinate values in context
- Graphing pattern relationships and describing what the graph shows
- A hierarchy of two-dimensional figures based on properties

LEARNING OBJECTIVES

1. Write an expression for a described calculation without evaluating it (e.g., 'add 8 and 7, then multiply by 2').
2. Evaluate a nested expression using correct order of operations.
3. Generate two patterns, form ordered pairs, and graph them on the coordinate plane.
4. Explain why every square is a rectangle but not every rectangle is a square.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Evaluates nested expressions correctly in 9 of 10 trials.
- Graphs ordered pairs accurately in 9 of 10 trials.
- Correctly places at least four quadrilateral types in a property hierarchy.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Coordinates are reversed: (3, 5) plotted as (5, 3).	Use a consistent verbal routine — 'across the hall, then up the stairs' — and require the student to say it aloud until it is automatic.
Order of operations is applied strictly left to right.	Grouping symbols and multiplication precedence must be practiced deliberately with contrasting pairs such as $2 + 3 \times 4$ and $(2 + 3) \times 4$.

TEACHING NOTE

This unit is a bridge to Grade 6 and to algebra. Writing an expression without evaluating it is genuinely new thinking for a fifth-grader and deserves explicit attention — it is the first step toward treating a symbol string as an object.

ASSESSMENT AND MASTERY POLICY

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ASSESSMENT	WHEN	WHAT IT MEASURES
Placement Assessment	Before enrollment	Weighted toward Grade 4 fraction and multi-digit operations; identifies gaps to close before Unit 2.
Unit Check	End of each unit · 30 min	Unit mastery criteria plus two extended reasoning items.
Fluency Probe	Every 2nd lesson · 3 min	Multiplication facts and multi-digit computation; continues all year.
Fraction Operations Diagnostic	After Units 2 and 3	The highest-stakes internal check of Grade 5; failure triggers targeted remediation before Unit 4.
Mid-Year Assessment	After Unit 3	Cumulative Units 1–3.
End-of-Course Assessment	After Unit 6	All course outcomes; determines Grade 6 placement and whether the student is ready for an accelerated middle school track.

Վարժարան reports mastery, not grades. Each topic in the student's profile is marked **Mastered**, **Developing**, or **Needs Review** against the criteria stated in each unit above. A topic is marked Mastered only when the criterion is met on two separate occasions at least one week apart — this prevents short-term recall from being recorded as durable learning. Topics marked Needs Review are automatically re-entered into homework and into the opening minutes of subsequent lessons until they are re-mastered.

PACING OPTIONS

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TRACK	CADENCE	EXPECTED COMPLETION
Standard	1 lesson/week, 60 min	Full course in approximately 45–50 weeks.
Accelerated	2 lessons/week	Full course in approximately 24–28 weeks; recommended for students targeting Algebra I in Grade 7 or 8.
Intensive / Catch-Up	2–3 lessons/week	18–22 weeks with diagnostic-driven ordering.
Summer Program	3 lessons/week, 6 weeks	Fraction and decimal operations intensive (Units 2–4) — the most requested summer program at this level.

Pacing is assigned after the placement assessment and reviewed at the mid-year assessment. A student who is meeting mastery criteria early may be moved to the accelerated track; a student who is not may be moved to intensive without any change in the course content itself.

HOMEWORK AND INDEPENDENT PRACTICE

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Thirty minutes per lesson: six to eight problems including at least two word problems and two cumulative review items. Beginning in Unit 2, every assignment requires the student to estimate before computing on at least one problem and to write the estimate down. Students who consistently finish in under fifteen minutes with full accuracy are flagged for the accelerated track.

PROGRESS MILESTONES REPORTED TO PARENTS

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These are the achievements the parent portal announces as the student reaches them. They replace attendance counts as the primary measure of progress.

- Understands place value through thousandths and powers of ten.
- Multiplies multi-digit whole numbers fluently and divides by two-digit divisors.
- Adds and subtracts fractions and mixed numbers with unlike denominators.
- Multiplies fractions by fractions and interprets multiplication as scaling.
- Divides with unit fractions and explains why the answers behave as they do.
- Performs all four operations with decimals to hundredths.
- Understands volume and finds volumes of prisms and composite solids.
- Writes and evaluates expressions with grouping symbols.
- Graphs points and pattern relationships on the coordinate plane.
- Completed Grade 5 Mathematics — elementary arithmetic complete; ready for Grade 6.

MATERIALS AND RESOURCES

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- Fraction strips, fraction circles, and area-model grids for fraction multiplication.
- Thousandths grids and place-value charts extending both directions from the decimal point.
- Unit cubes for volume (physical set at home strongly recommended).
- Coordinate grid paper.
- Վարժարան Grade 5 problem bank — approximately 1,200 items, with a 300-item fraction and decimal operations sub-bank.

Վարժարան · Varzharan — Exceptional Armenian educators. American academic standards. Personalized education. | This syllabus defines the standard course. Every enrolled student receives a personalized version of it, sequenced from their placement assessment and revised as their progress profile changes.